

SENTINEL AI — DEEPPFAKE INVESTIGATION REPORT

INVESTIGATION ID: CASE-TEST-M7 — DIGITAL EVIDENCE EXAMINATION

1. INVESTIGATION METADATA

|                       |                                       |                  |                              |
|-----------------------|---------------------------------------|------------------|------------------------------|
| Investigation ID:     | CASE-TEST-M7                          | Case Status:     | CASE_UNDER_INVESTIGATION     |
| Case Title:           | Milestone 7 Frontend Integration Test | Evidence Items:  | 1                            |
| Investigator / Owner: | Test User                             | Assigned Expert: | Test User                    |
| Incident Date:        | N/A                                   | Analysis Date:   | 2026-09-25 06:34 UTC         |
| Created Date:         | 2026-09-24 21:45 UTC                  | Model Version:   | Sentinel AI V1.7-A Dual-Head |

2. EXECUTIVE SUMMARY

|                         |                          |                           |                   |
|-------------------------|--------------------------|---------------------------|-------------------|
| Total Evidence Analyzed | Tampered Classifications | Authentic Classifications | Localization Maps |
| 1                       | 1                        | 0                         | 1 / 1             |

3. DETAILED EVIDENCE EXAMINATION

EVIDENCE ITEM E-001: evidence\_m7\_test.jpg

|                   |                                                                            |                     |                        |
|-------------------|----------------------------------------------------------------------------|---------------------|------------------------|
| Evidence ID:      | 14                                                                         | Uploaded By:        | Test User (Case Owner) |
| MIME Type:        | image/jpeg                                                                 | File Size:          | 41.20 KB               |
| Upload Timestamp: | 2026-09-25 06:34 UTC                                                       | Analysis Timestamp: | 2026-09-25 06:34 UTC   |
| SHA-256 Hash:     | 57244f424f284ffbb6aae62425f06ea5000000<br>00000000000000000000000000000000 |                     |                        |



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

| Metric                | Sentinel AI V1.7-A Value     | Threshold Criteria     |
|-----------------------|------------------------------|------------------------|
| Classification Result | TAMPERED / MANIPULATED MEDIA | Threshold: 0.40        |
| Tampered Probability  | 75.0%                        | Range: [0.0% - 100.0%] |
| Authentic Probability | 25.0%                        | Range: [0.0% - 100.0%] |
| Spatial Localization  | Heatmap Artifact Generated   | Threshold: 0.35        |

## 4. MODEL METHODOLOGY & TECHNICAL DETAILS

**Architecture Overview:** Sentinel AI V1.7-A is a frozen dual-head deep learning architecture optimized for image deepfake detection and pixel-level spatial localization.

**Preprocessing Stream:** Input images are normalized and passed through a dual-stream RGB backbone (EfficientNet-B0) paired with a high-pass Laplacian residual filter bank capturing high-frequency forensic noise signatures.

**Dual Heads:**

- *Classification Head:* Computes softmax probability across authentic vs. tampered classes using a calibrated decision threshold of **0.40**.
- *Localization Head:* Generates pixel-level feature heatmaps highlighting manipulated regions using a spatial activation threshold of **0.35**.

## 5. FORENSIC INTERPRETATION & DISCLAIMER

**OFFICIAL FORENSIC DISCLAIMER**

This report presents automated deepfake classification and spatial localization results produced by Sentinel AI V1.7-A. These findings provide probabilistic indicators based on forensic noise, boundary inconsistencies, and facial feature distributions. This document is intended to assist law enforcement, legal experts, and digital forensic investigators as corroborative evidence. Final evidentiary conclusions should incorporate qualified human expert examination.